Claims Description Normalizer

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1. Project Description

The **Claims Description Normalizer** is an AI-powered NLP system designed to convert **unstructured, free-text insurance claim descriptions** written by customers or adjusters into **clean, structured, machine-readable data**.

In real-world insurance systems, claim descriptions are often inconsistent, verbose, and ambiguous. This project uses **Natural Language Processing (NLP)** and **Large Language Models (LLMs)** to automatically extract key claim attributes such as **loss type, severity, affected asset, location, and incident date**, and outputs them in a standardized **JSON format**.

This enables faster claim triaging, improved automation, and reduced manual intervention in insurance workflows.

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2. Problem Statement

Insurance companies receive thousands of claims daily in **free-text form**, such as emails, app inputs, or handwritten notes digitized into text. These descriptions:

- Lack a fixed format
- Vary by user language and writing style
- Contain incomplete or noisy information
- Require manual reading and classification

Challenges:

- Manual processing is **time-consuming and error-prone**
- Inconsistent data impacts **analytics and fraud detection**
- Traditional rule-based systems fail on complex or vague text
- Delays in claim categorization increase settlement time

👉 **Hence, there is a need for an intelligent system that can automatically normalize raw claim text into structured data.**

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3. Objectives

The key objectives of this project are:

1. To design an NLP system that understands **unstructured insurance claim text**
2. To automatically identify:
    - Type of loss
    - Severity of damage

    - Affected asset
    - Incident location and date
3. To convert free-text input into **standardized JSON output**
4. To reduce manual effort in claim intake processing
5. To build a scalable and deployable solution using modern AI tools
6. To provide a clean UI for easy interaction and testing

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4. Proposed Solution

The proposed solution is an **AI-driven claims normalization pipeline** consisting of:

- **Text preprocessing** to clean and standardize input
- **Entity extraction** using spaCy NLP
- **Prompt-based data structuring** using Gemini LLM
- **Structured JSON generation** for downstream systems

Key Idea:

Instead of hardcoding rules, the system uses **LLMs** trained on large corpora to understand context and semantics, making it robust to spelling errors, informal language, and incomplete descriptions.

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5. Tech Stack

Frontend

- **Streamlit**
  Used to create a fast, interactive web UI for claim input and result visualization.

Backend / NLP

- **Python** – Core programming language
- **spaCy** – Named Entity Recognition (NER) for extracting locations, dates, organizations, etc.
- **Regular Expressions (re)** – Text cleaning and normalization

LLM / AI

- **Google Gemini 2.5 Flash-Lite**
    - Prompt-based information extraction
    - Converts raw text into structured JSON
    - Fast and cost-efficient for real-time use

Environment & Tools

- **python-dotenv** – Secure API key management

- **JSON** – Structured output format

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6. Workflow of the Project

1. **User inputs claim description** via UI
2. **Text preprocessing**
    - Remove extra spaces
    - Normalize formatting
3. **spaCy NLP processing**
    - Sentence parsing
    - Named Entity Recognition (date, location, etc.)
4. **LLM-based normalization**
    - Prompt sent to Gemini model
    - Model extracts structured attributes
5. **JSON validation**
    - Ensure output is valid and structured
6. **Result display**
    - Structured claim JSON
    - Extracted entities for transparency

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7. Input and Output Parameters

Input

Parameter	Description
claim_text	Raw, unstructured claim description

Output (JSON)

Field	Description
loss_type	Type of loss (accident, theft, fire, etc.)
severity	Damage severity (low / medium / high)
affected_asset	Asset involved (car, house, phone, etc.)
incident_date	Date of incident (if available)

Field	Description
location	Incident location (if available)
short_summary	One-line claim summary

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8. Uniqueness of the Project

- 🔹 **Prompt-based structuring instead of rigid rules**
- 🔹 Handles vague and noisy real-world text
- 🔹 Combines **traditional NLP + LLM intelligence**
- 🔹 Outputs directly usable JSON for APIs
- 🔹 Faster claim triage and decision-making
- 🔹 Scalable across multiple insurance domains

Unlike traditional systems, this model **understands context**, not just keywords.

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9. Conclusion

The **Claims Description Normalizer** successfully demonstrates how modern NLP and LLM technologies can transform unstructured insurance data into structured, actionable insights.

By automating claim normalization:

- Manual effort is reduced
- Claim processing becomes faster
- Data quality improves significantly

This project bridges the gap between **human-written text** and **machine-readable insurance systems**, making it highly relevant for real-world deployment.

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10. Future Work

The system can be enhanced further by:

1. **Confidence scoring** for extracted fields
2. **Fraud risk indicators**
3. **Multilingual claim support**
4. **Batch processing via CSV uploads**
5. **Fine-tuned NER models for insurance-specific entities**
6. **Integration with claim management systems**

7. **Human-in-the-loop validation**
8. **Explainability dashboard for compliance**